Data Analytics
Exercise 3: SQL Case Statements

1. Classify each product by price:
- 'Expensive' if price > 1000
- 'Mid-range' if price between 100 and 1000
- 'Budge' if price < 100

```
SELECT product_name,
       price,
       CASE
         WHEN price > 1000 THEN 'Expensive'
         WHEN price BETWEEN 100 AND 1000 THEN 'Mid-range'
         ELSE 'Budget'
       END AS price_category
FROM   products;
```

Product_name	price	price_category
Laptop	1200.00	Expensive
phone	800.00	Mid-range
keyboard	45.00	Budget
Monitor	300.00	Mid-range
Mouse	25.00	Budget

2. Label each order:
- 'High-value' for orders ≥ 1000
- 'Medium-value' for orders between 500 - 999.99
- 'Low-value' for orders < 500

```
SELECT customer_name,
        amount,
        CASE WHEN amount ≥ 1000 THEN 'High-value'
            WHEN amount BETWEEN 500 AND 999.99 THEN 'Medium-value'
            ELSE 'Low-value'
        END AS order_value_category
FROM orders;
```

Customer_name	amount	order_value_category
Alice	150.00	Low-value
Bob	560.00	Medium-value
Charlie	999.99	Medium-value
Diana	45.50	Low-value
Ethan	1200.00	High-value

3. Categorize employee position:

- If in 'IT' and salary > 80000 → 'Senior IT'
- If in 'HR' and salary > 55000 → 'Experienced HR'
- Otherwise → 'Staff'

```
SELECT emp_name,
        department,
        Salary,
    CASE WHEN department = 'IT' AND Salary > 80000 THEN 'Senior IT'
        WHEN department = 'HR' AND Salary > 55000 THEN 'Experienced HR'
        ELSE 'Staff'
    END AS position_level
FROM employees;
```

emp_name	department	Salary	position_level
John	IT	85000	Senior IT
Sara	HR	60000	Experienced HR
Mark	IT	75000	Staff
Lucy	Finance	95000	Staff
Tom	HR	55000	Staff

4. Assign a letter grade.

- ≥ 90 : 'A'
- 80 - 89 : 'B'
- 70 - 79 : 'C'
- 60 - 69 : 'D'
- < 60 : 'F'

```
SELECT student_name,
        Score,
    CASE ~~WHEN~~
        WHEN Score ≥ 90 THEN 'A'
        WHEN Score BETWEEN 80 AND 89 THEN 'B'
        WHEN Score BETWEEN 70 AND 79 THEN 'C'
        WHEN Score BETWEEN 60 ~~69~~ AND 69 THEN 'D'
        ELSE 'F'
    END AS grade
From Students;
```

Student_name	Score	grade
Anna	92	A
Ben	76	C
Cara	59	F
David	83	B
Ella	68	D

5. Label delivery performance:

- ≤ 30 mins : 'Fast'
- 31 - 60 mins : 'On time'
- 60 mins : 'Late'

```
SELECT delivery_id,
        delivery_time_minutes,
        CASE
          WHEN delivery_time_minutes ≤ 30 mins THEN 'Fast'
          WHEN delivery_time_minutes BETWEEN 31 AND 60 MINS THEN 'On time'
          ELSE 'Late'
        END AS performance
FROM    deliveries;
```

delivery_id	delivery_time_minutes	Performance
1	45	On time
2	80	Late
3	30	Fast
4	65	Late
5	100	Late.

6. Convert priority to labels

- 3 → 'High'
- 2 → 'Medium'
- 1 → 'Low'

```
SELECT issue_type,
        Priority,
    CASE
      WHEN Priority = 3 THEN 'High'
      WHEN Priority = 2 THEN 'Medium'
      ELSE 'Low'
```

```
END AS priority_label
FROM tickets;
```

issue_type	priority	priority_label
Login issue	1	Low
Server down	3	High
Slow system	2	Medium
Email error	2	Medium
Password reset	1	Low

7. Calculate attendance % and classify
- ≥ 90% → 'Excellent'
- 75 - 89% → 'Good'
- < 75% → 'Needs improvement'

```
SELECT student_id,
        (days_present / total_days)* 100 AS attendance_percentage
    CASE WHEN (days_present/total_days) * 100 ≥ 90% THEN 'Excellent'
        WHEN (days_present/total_days)* 100 BETWEEN 75% AN 89% THEN 'Good'
        ELSE 'Needs improvement'
    END AS attendance_status
FROM attendance;
```

Student_id	attendance_percentage	attendance_status
1	90%	Excellent
2	60%	Needs improvement
3	96%	Excellent
4	50%	Needs improvement
5	100%	Excellent.

8. Label Stock Status

- 0 → 'Out of stock'
- 1-5 → 'Low stock'
- 5 → 'In Stock'

```
SELECT product_id,
       Stock_qty,
    CASE
      WHEN stock_qty = 0 THEN 'out of stock'
      WHEN stock_qty BETWEEN 1 AND 5 THEN 'LOW STOCK'
      ELSE 'in stock'
    END AS stock_status
FROM products_inventory;
```

product_id	stock_qty	Stock_status
1	5	Low stock
2	0	out of stock
3	25	In stock
4	10	In stock
5	3	Low stock

9. Classify by size

- ≥ 25 → 'Large'
- 10-24 → 'Medium'
- < 10 → 'small'

```
SELECT subject,
       enrolled_students,
    CASE
      WHEN enrolled_students ≥ 25 THEN 'Large'
      WHEN enrolled_students BETWEEN 10 AND 24 THEN 'Medium'
```

```
        ELSE 'Small'
    END AS class_size_category
FROM classes;
```

Subject	enrolled_students	Class_size_category
Math	30	Large
English	25	Large
Science	15	Medium
Art	5	Small
History	20	Medium

10. Apply discount flag:
- If payment_method = 'cash' and amount ≥ 200 → 'Eligible for Discount'
- Otherwise → 'Not eligible.

```
SELECT payment_id,
       amount,
       payment_method,
    CASE
      WHEN payment_method = 'cash' and amount ≥ 200 THEN 'Eligible for
      Discount'
      ELSE 'Not eligible'
    END AS discount_eligibility
From payments;
```

Payment_id	amount	payment_method	discount_eligibility
1	50.00	card	Not eligible
2	200.00	Cash	Eligible for Discount
3	150.00	Card	Not eligible
4	75.00	Paypal	Not eligible
5	300.00	Cash	Eligible for Discount